Yang Hu

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Education

Carnegie Mellon University, Pittsburgh, United States 08/2026–Present
Ph.D. in Computer Science
Advised by William Kuszmaul and Elaine Shi

Tsinghua University, Beijing, China 08/2022–06/2026
B.Eng. in Computer Science, Yao Class, Institute for Interdisciplinary Information Sciences

Research Visits

- **Carnegie Mellon University** Spring 2025
  Student intern advised by William Kuszmaul

Publications

Authors are in alphabetical order.

Iterative Randomized Rounding for Better-Than-2 k-Median Approximation
Jarosław Byrka, Yuhao Guo, **Yang Hu**, Shi Li, Yongho Shin, Phanu Vajanopath, Chengzhang Wan, Zaixuan Wang.

Quadratic Probing Insertions Are $\varepsilon^{-(1+o(1))}$ Time
Yang Hu, William Kuszmaul, Jingxun Liang, Stefan Walzer, Huacheng Yu, Renfei Zhou.

Optimal Time-Space Tradeoff for Dynamic Difference-Encoded Dictionaries
Guy E. Blelloch, **Yang Hu**, William Kuszmaul, Jingxun Liang, Renfei Zhou.
In *Proc. SODA*, 2027 (to appear).

Dynamic Entropy-Encoded Arrays in $O(1)$ Time with Nearly Optimal Space
Guy E. Blelloch, **Yang Hu**, William Kuszmaul, Tianxiao Li, Renfei Zhou.
In *Proc. FOCS*, 2026 (to appear).

k-Clustering via Iterative Randomized Rounding
Jarosław Byrka, Yuhao Guo, **Yang Hu**, Shi Li, Chengzhang Wan, Zaixuan Wang.
In *Proc. FOCS*, 2026 (to appear).

Quadratic Probing Revisited: Smoothed Analysis and the Fall of Robin Hood
Yang Hu, William Kuszmaul, Jingxun Liang, Stefan Walzer, Huacheng Yu, Renfei Zhou.

Fast Concurrent Primitives Despite Contention
Michael A. Bender, Guy E. Blelloch, Martín Farach-Colton, **Yang Hu**, Rob Johnson, Rotem Oshman, Renfei Zhou.
In *Proc. SPAA*, 2026.

New Algebrization Barriers to Circuit Lower Bounds via Communication Complexity of Missing-String
Lijie Chen, **Yang Hu**, Hanlin Ren.
In *Proc. ITCS*, 2026.

Nearly Optimal Bounds for Stochastic Online Sorting
Yang Hu.
In *Proc. SODA*, 2026.

Static Retrieval Revisited: To Optimality and Beyond
Yang Hu, William Kuszmaul, Jingxun Liang, Huacheng Yu, Junkai Zhang, Renfei Zhou.
In *Proc. FOCS*, 2025.

Nash Social Welfare with Submodular Valuations: Approximation Algorithms and Integrality Gaps
Xiaohui Bei, Yuda Feng, **Yang Hu**, Shi Li, Ruilong Zhang.
In *Proc. STOC*, 2026.

Constant Approximation for Weighted Nash Social Welfare with Submodular Valuations
Yuda Feng, **Yang Hu**, Shi Li, Ruilong Zhang.
In *Proc. STOC*, 2025.

Optimal Static Dictionary with Worst-Case Constant Query Time
Yang Hu, Jingxun Liang, Huacheng Yu, Junkai Zhang, Renfei Zhou.
In *Proc. STOC*, 2025.

Selected Awards and Scholarships

- **Yao Award (Silver Medal)** 2025
  Awarded to 3 students in Yao Class each year

Competitive Programming (Selected)

- **The 2022 ICPC Asia East Continent Final Contest** 2023
  Gold award, 4th place (with teammates Junkai Zhang and Hao Hu)
- **ACM International Collegiate Programming Contest, Asia Regional Contest, Nanjing Site** 2022
  Gold medal, 3rd place (solo)
- **ACM International Collegiate Programming Contest, Asia Regional Contest, Jinan Site** 2022
  Gold medal, 2nd place (with teammates Junkai Zhang and Hao Hu)
- **China National Olympiad in Informatics** 2021
  Gold medal, 3rd place
- **The Asia-Pacific Informatics Olympiad** 2021
  Ranked 4th in China
- **CCF NOI Winter Camp** 2021
  Gold medal, 1st place

Service

- **Algorithm Analysis and Design (Tsinghua University)** Autumn 2024
  Teaching assistant (Instructor: Ran Duan)

Skills

- TOEFL iBT Score 115 02/2024
- Japanese N1 06/2022